

Tide retrospective: *generic- k J-function asymptotic for monomial potentials*

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Abstract

For the single-monomial potential $K(w) = w^{2k}/(2k)!$ with $k \in \mathbb{N}$, $k \geq 1$, and a continuous, globally bounded prior π , we formalise the leading-order asymptotic of the J-function $J_k(t) = \int e^{-tK(w)} \pi(w) dw$ as $t \rightarrow \infty$. The headline constant is $\pi(0) \cdot \frac{1}{k} \cdot ((2k)!)^{1/(2k)} \cdot \Gamma(1/(2k))$. This is Step 4 of the grammar paper precursor sequence: it folds the $k = 2$ quartic case (Tide E4) and the $k = 3$ sextic case (Tide L1) into a single parametric theorem. New file `Laplace/OneD/JFunctionMonomial.lean`, 497 lines, three public theorems plus four private helpers; zero `sorry`, zero `axiom`, zero `native_decide`; single-session formalisation. The Step 2 deliberation with GPT-5.5 Pro converged in one round: both votes for the headline (full-form) candidate, with GPT contributing a small public-API trim (keep the integrability helper private) and a critical tactical note that `ring` cannot unify under generic exponents like $2*k$, forcing the use of `mul_pow` in the substitution lemma’s load-bearing step.

1 Setting

The grammar paper precursor sequence is a chain of formalised asymptotic-method results that ground the grammar paper’s Sections on resolution of singularities. Three earlier tides have laid the seabed:

- Tide 7 (the original *grammar precursor* tide, 2 May) established the substitute-then-DCT skeleton.
- Tide E4 (this morning) instantiated the J-function asymptotic at $k = 2$ in the quartic file,¹ with closed-form constant $\frac{1}{2} \cdot 24^{1/4} \cdot \Gamma(1/4)$ and proof template *substitute-then-DCT*.
- Tide L1 (immediately after E4) mirrored the proof at $k = 3$ in the sextic file,² with constant $\frac{1}{3} \cdot 720^{1/6} \cdot \Gamma(1/6)$.

The L1 retrospective explicitly named the generic- k abstraction as “*now ripe*”: with both $k = 2$ and $k = 3$ in the same shape, the abstraction over $K(w) = w^{2k}/(2k)!$ becomes the natural next step. The constant generalises cleanly to $\frac{1}{k} \cdot ((2k)!)^{1/(2k)} \cdot \Gamma(1/(2k))$, and the proof template generalises modulo cast bookkeeping.

The Step 2 deliberation between Claude and GPT-5.5 Pro picked the headline candidate (full asymptotic plus closed-form partition, three public theorems) over a thinner alternative that would have postponed the asymptotic corollary as a follow-up. Both models voted the same target in a single round; GPT additionally trimmed the public API (`kth_integrable_pow` kept private, since

¹`Laplace/OneD/JFunctionQuartic.lean`.

²`Laplace/OneD/JFunctionSextic.lean`.

only $n = 0 \in \mathbb{N}$ is needed inside this file) and reshaped the proof tactics to use `mul_pow` rather than the `ring` idiom from L1 (`ring` cannot unify under a parametric $2*k$).

The markdown tide `log3` records the full deliberation; the GPT consult is preserved at the sibling `gpt_responses_n1/n1_v1.md`.

2 The thing formalised

For $k \in \mathbb{N}$ with $k \geq 1$ and a continuous, globally bounded prior $\pi : \mathbb{R} \rightarrow \mathbb{R}$ (with global bound M),

$$\lim_{t \rightarrow \infty} t^{1/(2k)} \cdot \int_{\mathbb{R}} e^{-t w^{2k}/(2k)!} \pi(w) dw = \pi(0) \cdot \frac{1}{k} \cdot ((2k)!)^{1/(2k)} \cdot \Gamma\left(\frac{1}{2k}\right).$$

The Lean signature⁴:

```
theorem kth_jfunction_asymptotic
  {k : ℕ} (hk : 1 ≤ k)
  {prior : ℝ → ℝ} (hprior_cont : Continuous prior)
  {M : ℝ} (hprior_bd : x, |prior x| ≤ M) :
  Tendsto (fun t : ℝ =>
    t ^ ((1 : ℝ) / ((2 * k : ℕ) : ℝ)) *
    w : ℝ, Real.exp (-(t * w ^ (2 * k) /
      (Nat.factorial (2 * k) : ℝ)))
      * prior w)
    atTop
    ( (prior 0 * (1 / (k : ℝ))
      * ((Nat.factorial (2 * k) : ℝ))
        ^ ((1 : ℝ) / ((2 * k : ℕ) : ℝ))
      * Real.Gamma ((1 : ℝ) / ((2 * k : ℕ) : ℝ))))
  '''
```

`\noindent` The two existing files (`\texttt{JFunctionQuartic.lean}` for $k=2$, `\texttt{JFunctionSextic.lean}` for $k=3$) are not yet refactored as 5-line specialisations of this theorem, since both are on unmerged branches at the base time; that is a follow-up tide once all three land on `\texttt{main}`.

The closed-form partition is also exported:

```
\begin{verbatim}
theorem kth_partition {k : ℕ} (hk : 1 ≤ k) {t : ℝ} (ht : 0 < t) :
  x : ℝ, Real.exp (-(t * x ^ (2 * k) /
    (Nat.factorial (2 * k) : ℝ)))
    = (1 / (k : ℝ))
      * ((Nat.factorial (2 * k) : ℝ) / t)
        ^ ((1 : ℝ) / ((2 * k : ℕ) : ℝ))
\end{verbatim}
```

³In the SRI repo at `projects/primer/tide-log/`, file `2026-05-07-tide-generic-k-j-function.md`.

⁴`Laplace/OneD/JFunctionMonomial.lean`, `kth_jfunction.asymptotic`.

`* Real.Gamma ((1 : ℝ) / ((2 * k : ℕ) : ℝ))`

and the centred DCT core `kth_jfunction_centered_tendsto_zero`, which is the load-bearing intermediate.

The hypothesis $k \geq 1$ is mathematically essential: at $k = 0$ the “potential” $w^0/0! = 1$ is constant, making $e^{-t} \pi(w)$ have no concentration behaviour and the asymptotic trivial in the wrong way. Continuous-and-bounded on the prior is the slightly stronger form taken in E4/L1 over the deliberation’s `AEMeasurable + ContinuousAt 0`; weakening across all three files is a separate follow-up (E4 follow-up L3, now ripe across two files).

3 Proof strategy

The proof is substitute-then-DCT, identical in structure to E4/L1 modulo cast bookkeeping. There are four pieces.

The substitution. Set $a := t^{1/(2k)}$, so $a^{2k} = t$. The change of variables $u = a \cdot w$ via Mathlib’s `Measure.integral_comp_mul_right` reshapes

$$t^{1/(2k)} \int_{\mathbb{R}} e^{-tw^{2k}/(2k)!} (\pi(w) - \pi(0)) dw$$

into

$$\int_{\mathbb{R}} e^{-u^{2k}/(2k)!} (\pi(u/a) - \pi(0)) du.$$

The Jacobian bookkeeping is where most of the proof’s lines go. Concretely: $a^{2k} = t$ is proved by routing through `Real.rpow_natCast`, then `Real.rpow_mul`, then `Real.rpow_one`, with the intermediate identity $\alpha \cdot p_{\mathbb{R}} = 1$ (where $\alpha := 1/p_{\mathbb{R}}$ and $p_{\mathbb{R}} := ((2k)_{\mathbb{N}})_{\mathbb{R}}$) factored out as a named `have` so the calc chain stays linear.

The load-bearing line is the pointwise identity $(w \cdot a)^p = t \cdot w^p$. In E4 and L1, this was closed by `ring` after specialising to fixed $p \in \{4, 6\}$. For generic $p = 2k$, `ring` cannot unify the exponents and fails. The fix is `mul_pow`: $(w \cdot a)^p = w^p \cdot a^p$ unconditionally, then $a^p = t$ via `ha_p`, then `ring` to commute the multiplication. This was the single most important tactical correction GPT named at deliberation; without it the substitution lemma does not close.

The dominator’s integrability. The DCT envelope is $2M \cdot e^{-u^{2k}/(2k)!}$, which is t -independent (the substitution moved the t -dependence into the prior factor). For generic $k \geq 1$, integrability is established by Gaussian comparison via the universal pointwise inequality

$$x^2 \leq 1 + x^{2k} \quad (k \geq 1),$$

proved by case-split on $|x| \leq 1$ (where $x^2 \leq 1$ and $x^{2k} \geq 0$) versus $|x| > 1$ (where $x^{2k} \geq x^2$ since $2k \geq 2$). This gives $\frac{t}{(2k)!} \cdot x^2 - \frac{t}{(2k)!} \leq \frac{t \cdot x^{2k}}{(2k)!}$, hence $e^{-tx^{2k}/(2k)!} \leq e^{t/(2k)!} \cdot e^{-(t/(2k)!)\cdot x^2}$, the dominator being a Gaussian integrable by Mathlib’s `integrable_exp_neg_mul_sq` at $b = t/(2k)!$.

The centred DCT. The substituted integrand $F(t, u) := e^{-u^{2k}/(2k)!} \cdot (\pi(u/t^{1/(2k)}) - \pi(0))$ has a t -independent envelope $2M \cdot e^{-u^{2k}/(2k)!}$ (integrable from the previous step), and its pointwise limit is zero: as $t \rightarrow \infty$, $u/t^{1/(2k)} \rightarrow 0$ (the `tendsto_rpow_atTop` step at exponent $1/(2k) > 0$), and continuity of π at 0 gives $\pi(u/t^{1/(2k)}) - \pi(0) \rightarrow 0$. Mathlib’s filter version of the dominated convergence theorem⁵ closes.

⁵`tendsto_integral_filter_of_dominated_convergence`.

The closed-form partition. The partition $Z_k(t) = \int e^{-tw^{2k}/(2k)!} dw$ has even integrand, so it equals $2 \cdot \int_{(0,\infty)} e^{-tw^{2k}/(2k)!} dw$ via `integral_comp_abs`. The half-line integral is given by Mathlib’s master half-line moment formula⁶ at $q = 0$, $p = 2k$, $b = t/(2k)!$, yielding $(t/(2k)!)^{-1/(2k)} \cdot (1/(2k)) \cdot \Gamma(1/(2k))$. Doubling gives $(t/(2k)!)^{-1/(2k)} \cdot (2/(2k)) \cdot \Gamma(1/(2k))$, and $2/(2k) = 1/k$. The exponent flip $(t/(2k)!)^{-1/(2k)} = ((2k)!/t)^{1/(2k)}$ uses `inv_rpow` and `Real.rpow_neg`.

The asymptotic corollary. Splits $\pi(w) = (\pi(w) - \pi(0)) + \pi(0)$ inside the integrand. The centred piece tends to 0 (by the centred DCT theorem); the constant piece evaluates via `kth_partition` to $\pi(0) \cdot \frac{1}{k} \cdot ((2k)!/t)^{1/(2k)} \cdot \Gamma(1/(2k))$; multiplying by $t^{1/(2k)}$ and using $t^{1/(2k)} \cdot ((2k)!/t)^{1/(2k)} = ((2k)!)^{1/(2k)}$ closes.

4 Roadblocks and resolutions

This was a single-session formalisation with one substantive roadblock cluster, all related to the change in tactical class between fixed-exponent and parametric-exponent proofs.

Cast thrashing in the substitution lemma. The L1 proof of $a^p = t$ at fixed $p = 6$ was a calc chain of about ten lines mixing `Real.rpow_natCast`, `Real.rpow_mul`, and a final `norm_num`. Generic- k added a layer: $\alpha \cdot p_{\mathbb{R}} = 1$ now requires unfolding the `set` abbreviations (`hα_def`, `hpR_def`) so that `field.simp` sees the underlying expressions. The fix was to factor $\alpha \cdot p = 1$ out as a named `have` before the calc chain, with explicit `rw [hα_def, hpR_def]` to bring the unfolded form into scope.

GPT-5.5 Pro: For `a^(2*k) = t`, avoid rewriting to `2 * (k : ℝ)` unless you must. The clean route is: prove `a ^ p = t` via `Real.rpow_natCast`, then `Real.rpow_mul` with `α * pR = 1` factored out as a separate `have`.

ring cannot unify under generic exponents. The load-bearing line in the substitution lemma is $(w \cdot a)^p = t \cdot w^p$. In E4/L1, both files closed this by `ring` after specialising p to 4 or 6. For generic $p = 2k$, `ring` cannot unify the exponents — it sees p as an opaque natural number — and fails with a generic “unsolved goal”.

GPT-5.5 Pro: In the substitution lemma, use `mul_pow`, not `ring`. This matters. Fixed exponents like 4 and 6 were happy with `ring`, but generic $(2*k)$ won’t be.

The fix is one line: `rw [mul_pow, ha_p]; ring`. `mul_pow` reshapes $(w \cdot a)^p$ to $w^p \cdot a^p$ unconditionally; `ha_p` replaces a^p with t ; `ring` closes the resulting $w^p \cdot t = t \cdot w^p$.

set versus exact_mod_cast. After `set fac := (Nat.factorial p : ℝ)`, proving $0 < fac$ via a direct `exact_mod_cast Nat.factorial_pos` fails: the `cast` tactic sees `fac` as opaque and does not unfold the `set`. The fix is to first `change` the goal to the underlying form $(0 : ℝ) < (Nat.factorial p : ℝ)$, then dispatch with the `cast`. Three sites needed this fix; all minor but each blocked progress for ~ 2 min while diagnosed.

⁶`integral_rpow_mul_exp_neg_mul_rpow`.

Negative-exponent rewriting. Mathlib’s `integral_rpow_mul_exp_neg_mul_rpow` at $q = 0$ produces the exponent $-(0+1)/p_{\mathbb{R}}$, parsed as $-(0+1))/p_{\mathbb{R}} = -(1/p_{\mathbb{R}})$ by Lean’s arithmetic-precedence rules. The single `hexp_one` rewrite covers the positive form $(0 + 1)/p_{\mathbb{R}} = 1/p_{\mathbb{R}}$ that appears in the Γ argument, but does *not* fire on the negative form because the syntactic structure is different. Adding a parallel `hexp_neg` for the negated form fixed it.

set-defeq vs. convert cycles. The asymptotic theorem’s final goal massage initially used `convert ... using 1; funext t; rw [hp_eq]` chains to bridge the abbreviated form (using `p`, α , `fac` from `set`) and the public-form goal (using `2*k`, the explicit casts, the explicit factorial). Multiple “no goals to be solved” errors surfaced because `set` makes the abbreviated forms *definitionally equal* to the public ones, and `convert` closed the goal at a level above where I expected. The fix was to rebuild the centred-piece `Tendsto` with the `set` abbreviations directly in its statement, then use `kth_jfunction_centered_tendsto_zero without conversion` (Lean accepts it via `defeq`), then the final `Tendsto.congr`’ step closes the public-form goal also via `defeq`. Net: about 25 lines of `convert/funext` collapsed to 6 lines.

5 What was Mathlib, what was new

Mathlib provided. The full asymptotic stack:

- the master half-line moment formula (`integral_rpow_mul_exp_neg_mul_rpow`);
- the even-function full-line doubling lemma (`integral_comp_abs`);
- the substitution-by-constant integral lemma (`Measure.integral_comp_mul_right`);
- the filter version of dominated convergence⁷;
- the `rpow-tends-to-infinity` step (`tendsto_rpow_atTop`);
- integrability of the Gaussian dominator (`integrable_exp_neg_mul_sq`).

The Γ -function side is `Real.Gamma` from `Mathlib.Analysis.SpecialFunctions.Gamma.Basic`.

What was new. Three things, in increasing order of likely future reuse:

- `sq_le_one_add_pow_two_mul`: the universal pointwise inequality $x^2 \leq 1 + x^{2k}$ for $k \geq 1$. Cleaner and more general than the L1 file’s bespoke `sq_le_one_add_pow_six` or the Quartic file’s square-completion identity. Likely candidate for promotion to `lean-common` once a second use materialises.
- `kth_integrable`: full-line integrability of $e^{-tx^{2k}/(2k)!}$ for $k \geq 1$, via the universal Gaussian comparison above. Generalises and unifies the existing fixed-exponent integrability helpers in the quartic and sextic files at $n = 0$.
- `kth_partition`: closed-form full-line monomial partition function. Generalises the existing fixed-exponent partitions for the quartic and sextic potentials (which remain valid as standalone theorems on the existing branches; once E4 and L1 merge to `main`, they become 5-line specialisations, and the cleanup pass is a follow-up tide).

The substitution lemma `kth_jfunction_centered_subst` and the centred DCT theorem are private and bespoke; they would not add value to `lean-common` without further generalisation (e.g. to *any* potential admitting a power-law substitution).

⁷`tendsto_integral_filter_of_dominated_convergence`.

6 Lessons

- *For `laplace/CLAUDE.md`:* `ring` cannot unify exponents involving free natural-number variables. For $(w \cdot a)^p$ with parametric p , use `mul_pow` explicitly, not `ring`. The quartic and sextic files use `ring` as a fixed-exponent special case; generic- p proofs must replace it.
- *For `laplace/CLAUDE.md`:* `exact_mod_cast` does *not* unfold `set` abbreviations. After `set fac := (Nat.factorial p : ℝ)`, prove $0 < \text{fac}$ with an explicit `change` to the underlying form before the cast. (Already in `CLAUDE.md` as a related gotcha for cast directions; this run extends it to the `set-defeq` case.)
- *For `laplace/CLAUDE.md`:* when a Mathlib lemma produces an exponent of the form $-(q+1)/p$ in the conclusion (as `integral_rpow_mul_exp_neg_mul_rpow` does), Lean parses it as $(-(q+1))/p$ — distinct from $-((q+1)/p)$ for rewriting purposes. Provide *both* the positive-form rewrite (for the Γ argument) and the negative-form rewrite (for the `rpow` exponent) when massaging.
- *For the tide skill:* the deliberation transcript carried forward all three load-bearing tactical points (`mul_pow`, the `set/change` pattern, the explicit-cast preference) into the proof without any mid-proof thrashing-rescue consult needed. This is the second tide in a row (Tide I4 was the first) where the GPT-5.5 Pro consult at Step 2 contained *enough* tactical detail that Step 3 was mechanical Lean idiom. The pattern: when Step 2 has already named the Mathlib idiom and the specific friction points, the proof writes itself.
- *For tide-pick:* the L1 retrospective named N1 as “now ripe” *within hours* of L1 landing. Tide-pick picked it as the top recommendation in the v3 survey within another hour. The pattern of “retrospective Follow-ups become next-tide candidates” continues to be the dominant source of high-value tide steps; the strict-improvement-of-the-day pipeline is producing single-session formalisations at sub-day cadence.

7 Follow-ups

- *Refactor `JFunctionQuartic.lean` and `JFunctionSextic.lean` to be 5-line specialisations of `kth_jfunction_asymptotic`.* Cannot be done in this Tide because both files are on unmerged branches at base time. After E4, L1, and N1 all land on `main`, this is a focused cleanup pass: each existing file shrinks from ~ 280 lines to ~ 30 lines (the specialisation plus constant unification for the user-facing theorem name). One follow-up tide for the pair.
- *Hypothesis weakening across all three files.* The deliberation’s mathematically-minimal hypothesis class is `AEMeasurable + ContinuousAt 0 + global bound`; all three files use the slightly stronger `Continuous prior + global bound`. Now that the parametric form exists, weakening should land in N1 (`kth_*`) only — the specialisations inherit. Expected ~ 50 lines of `comp_quasiMeasurePreserving` plumbing.
- *Promote `sq_le_one_add_pow_two_mul` to `lean-common`.* Genuinely general; subsumes the L1 file’s bespoke `sq_le_one_add_pow_six`. Defer until a second downstream use (probably the hypothesis-weakening tide above).
- *Promote `kth_partition` to `lean-common`.* The natural full-line monomial partition formula; clean addition. Defer until a second use; the most likely trigger is the bounded-prior-monomial generalisation tide (B2 in the survey), which would consume it as primitive infrastructure.

- *Asymptotic-equivalence wrapper* $J_k(t) \sim \pi(0) \cdot Z_k(t)$ (*E4 follow-up L2, generalised*). A 15-line corollary that takes the `Tendsto` ($\mathfrak{t}^{1/(2k)} \cdot J_k$) form and divides through by `kth_partition`. Now applies to all $k \geq 1$ in one shot rather than once per file. Smallest candidate on the board going forward.
- `\leanref`-tag the grammar paper. Once the grammar paper has TeX statements pinning all three $k = 2, 3$, generic- k J-function asymptotics, tag with `\leanref` pinned to commit `ddb1f14` (this tide). Process tide; depends on paper-side draft.

Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2, including the three tactical points that made the proof single-session: the `mul_pow` recipe, the $\alpha \cdot p = 1$ factoring pattern, and the explicit-cast preference for the public theorem statement. The full consult is preserved at `projects/primer/tide-log/gpt.responses_n1/n1_v1.md`. No mid-proof thrashing-rescue consult was needed; the Step 2 advisory was sufficient.